

DS806.01

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Supply Chain

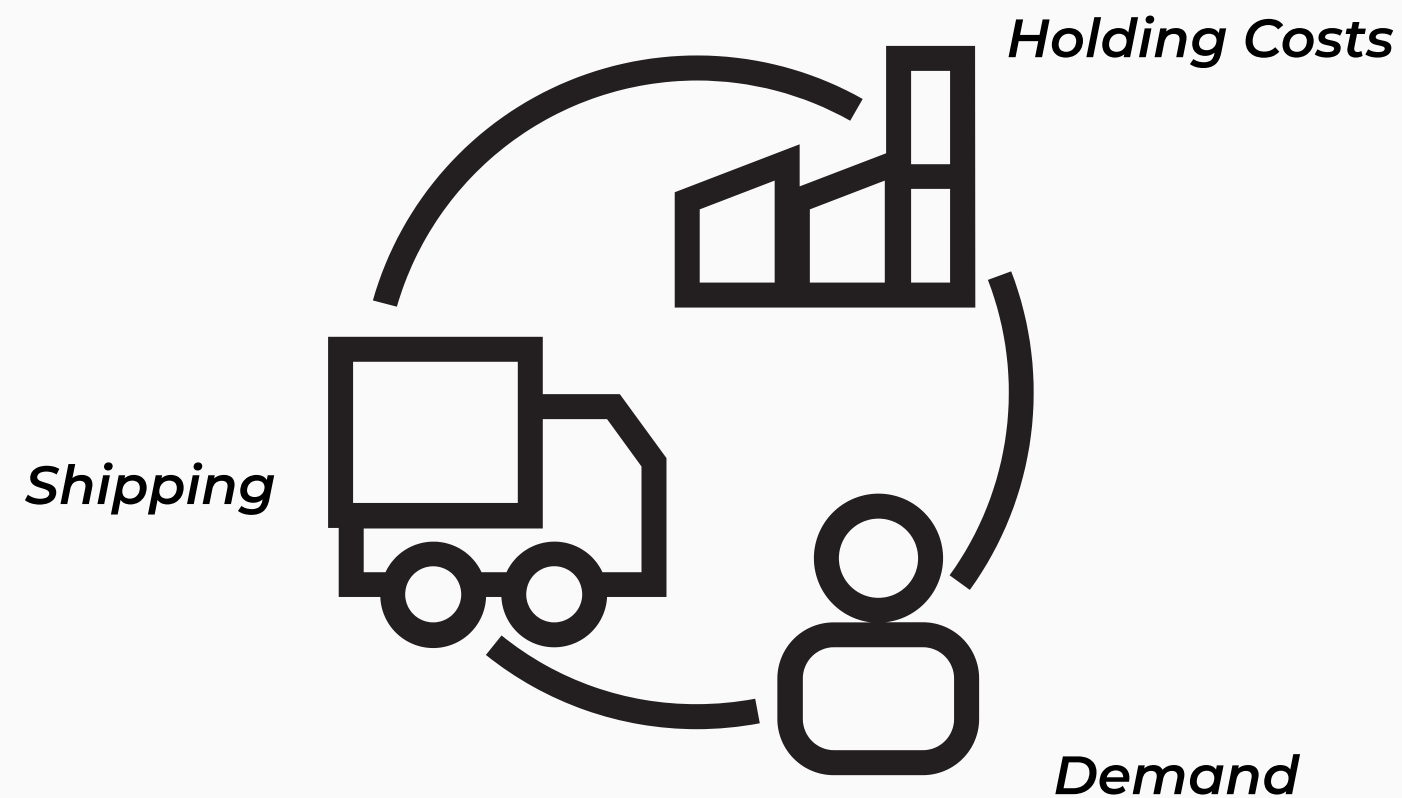
LOGISTIC PROBLEM



Content Summary

- *Introduction*
- *Problem definition*
- *Model walkthrough*
- *Analysis*
- *Results*
- *Major findings*
- *Lessons learned*
- *Conclusion*

Problem Definition

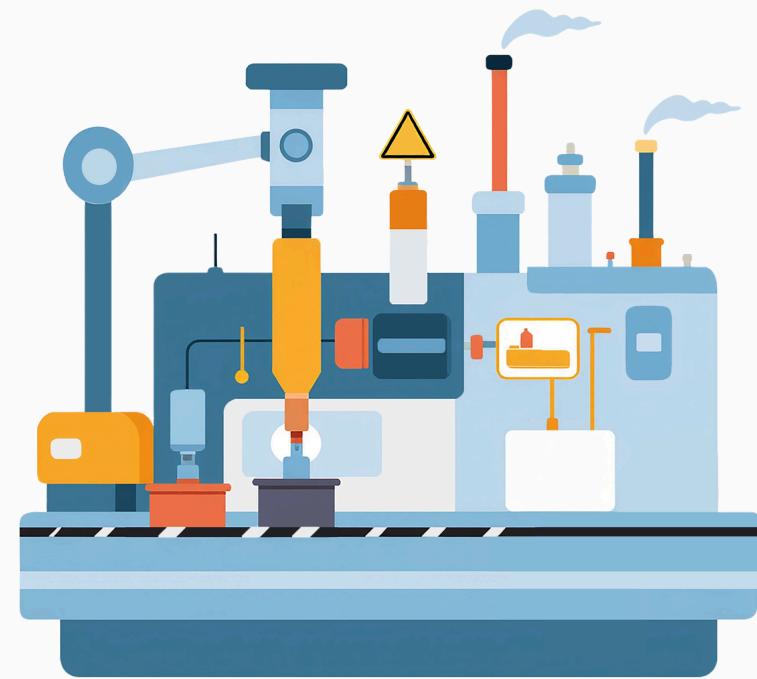


***HOW MANY DAYS TO
MEET DEMAND ONCE
DEMAND IS MAXIMIZED?***

19 Plants (Producers)

10 Ports (Shippers)

Model Flow



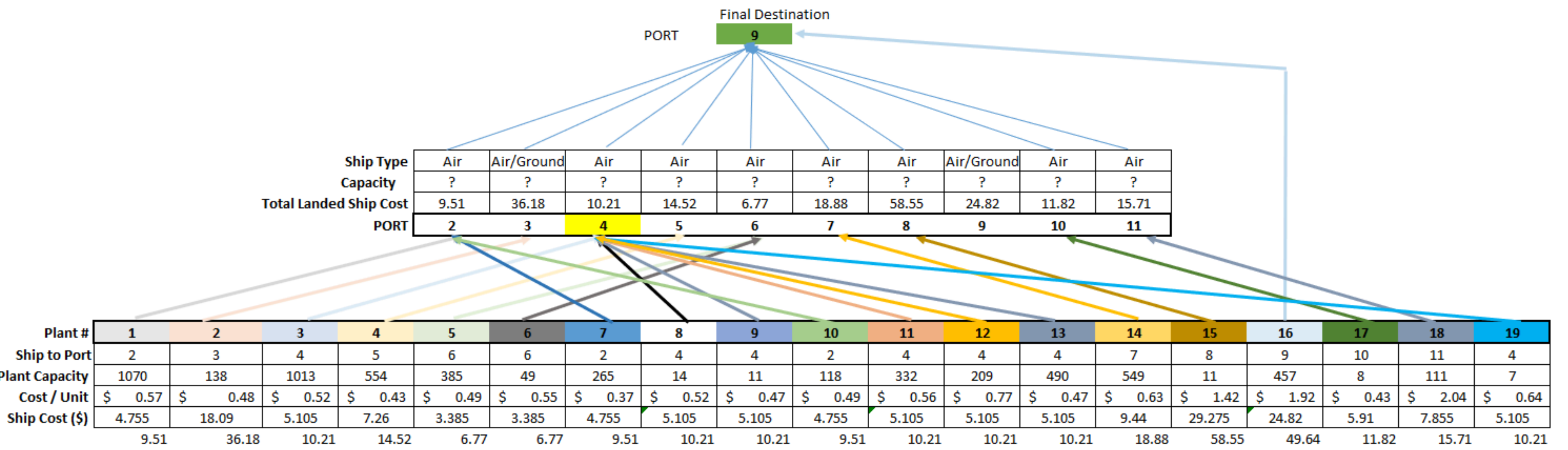
Plants (1-19)



Ports (2-11)



Mapped Out



! **Bottleneck** !

7 Different Plants ship to port 4

Constraints

- *All units must ultimately be shipped to Port 09*
- *Total units produced by each plant must be less than or equal to the plant's capacity*
- *Amount of units sent to each holding port must be less than or equal to the port's capacity*
- *Total units produced overall must not exceed the capacity of Port 09*

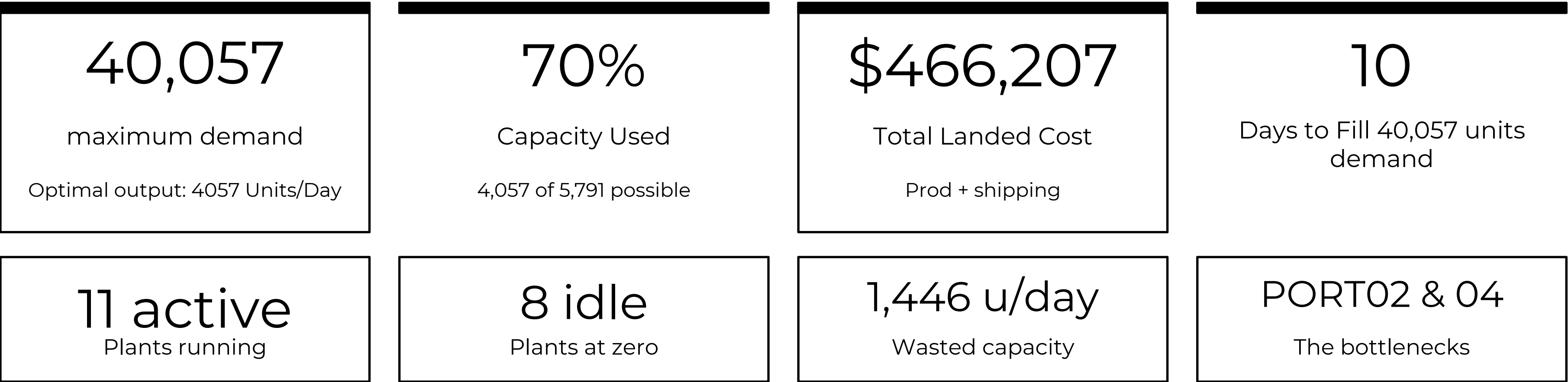


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Setup in Excel

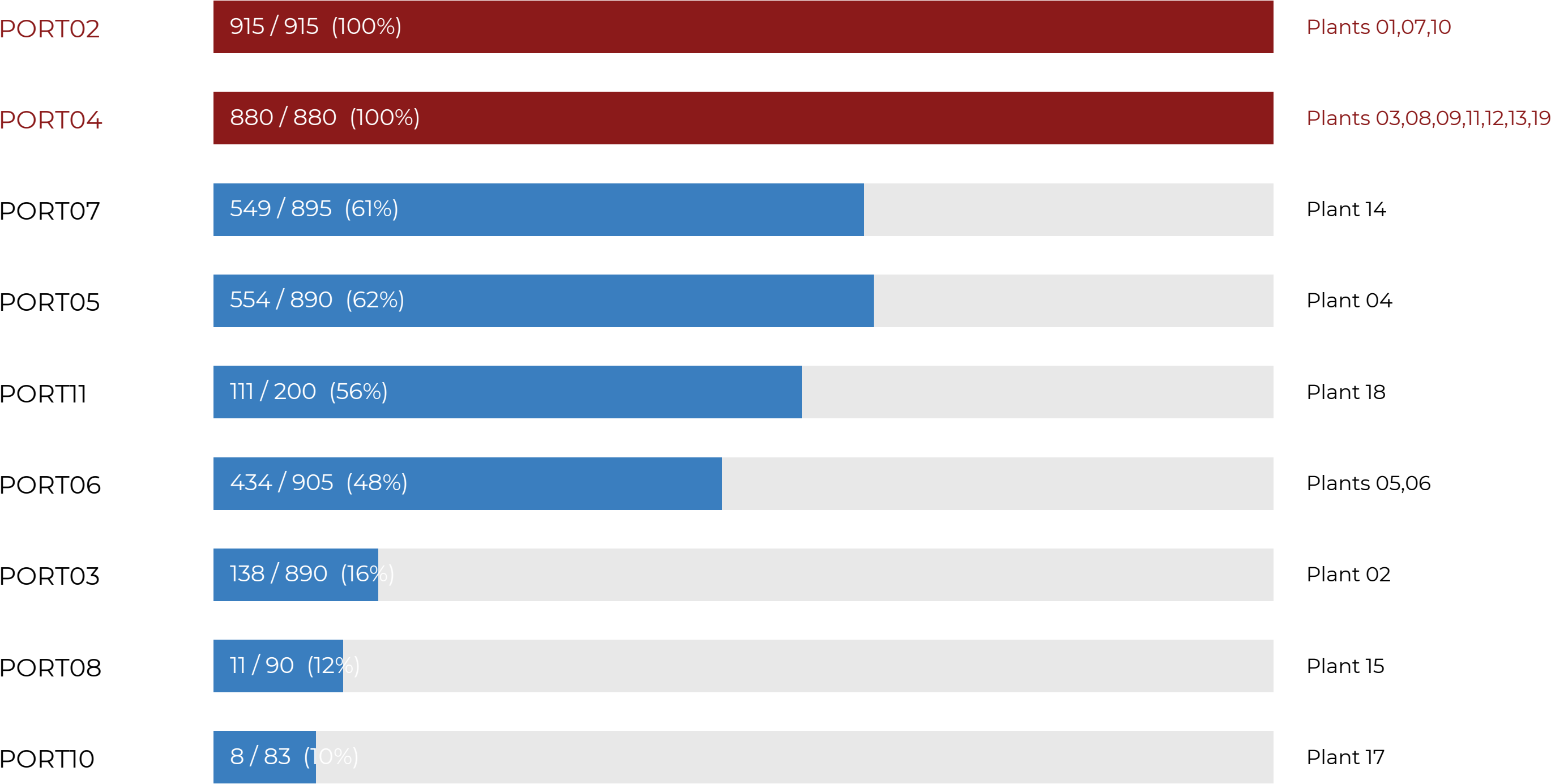
plant	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19								
Units produced	915	138	880	554	385	49	0	0	0	0	0	0	0	549	11	457	8	111	0	0	MAX	4057	40057				
																						0.10128	10	Days to meet demand			
Prod Cost	0.57	0.48	0.52	0.43	0.49	0.55	0.37	0.52	0.47	0.49	0.56	0.77	0.47	0.63	1.42	1.92	0.43	2.04	0.64		\$ 296,317.76						
Ship Cost per 100	9.51	36.18	10.21	14.52	6.77	6.77	9.51	10.21	10.21	9.51	10.21	10.21	10.21	18.88	58.55	49.64	11.82	15.71	10.21								
Ship Cost per unit	0.0951	0.3618	0.1021	0.1452	0.0677	0.0677	0.0951	0.1021	0.1021	0.0951	0.1021	0.1021	0.1021	0.1888	0.5855	0.4964	0.1182	0.1571	0.1021		\$ 69,194.57						
																				TC to port	\$ 365,512.33						
																				TC to port 9	\$ 466,207.07						
Plant Capacities																											
1	1																					915 <=	1070	1.07 M			
2		1																				138 <=	138				
3			1																			880 <=	1013				
4				1																		554 <=	554				
5					1																	385 <=	385				
6						1																49 <=	49				
7							1															0 <=	265				
8								1														0 <=	14				
9									1													0 <=	11				
10										1												0 <=	118				
11											1											0 <=	332				
12												1										0 <=	209				
13													1									0 <=	490				
14														1								549 <=	549				
15															1							11 <=	11				
16																1						457 <=	457				
17																	1					8 <=	8				
18																		1				111 <=	111				
19																			1			0 <=	7				
Total Capacities	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1		4057 <=	5791					
Port Capacities																											
2	1						1			1												915 <=	915	in millions, 9.15 million			
3		1																				138 <=	890				
4			1					1	1		1	1	1							1		880 <=	880				
5				1																		554 <=	890				
6					1	1																434 <=	905				
7														1								549 <=	895				
8															1							11 <=	90				
10																	1					8 <=	83				
11																		1				111 <=	200				
																						0	0				

What Did the Solver Find?



Key Insight: The network operates at only 70% of its theoretical capacity not because of plant limits, but because PORT02 and PORT04 are both using 100% capacity leaving no space for 1,446 units/day across 8 plants.

The Port Bottleneck and What It's Costing Us



THE PARADOX

PLANT07

\$0.37/unit

Cheapest plant in the network

PRODUCES ZERO UNITS

PORT02 IS FULL

If PORT02 expanded to activate PLANT07: +265 units/day and \$0.196/unit cost saving on that route of approximately \$52 saved every day.

Excel Sensitivity Report

Variable Cells (Units Produced per Plant)

Constraints (Port & Plant Capacities)

Microsoft Excel 16.103 Sensitivity Report
Worksheet: [Supply chain logisitcs problem (2).xlsx]Formulation
Report Created: 3/7/26 10:48:00

Variable Cells						
Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
\$B\$2	Units produced	915	0	1	1E+30	0
\$C\$2	Units produced	138	0	1	1E+30	1
\$D\$2	Units produced	880	0	1	1E+30	0
\$E\$2	Units produced	554	0	1	1E+30	1
\$F\$2	Units produced	385	0	1	1E+30	1
\$G\$2	Units produced	49	0	1	1E+30	1
\$H\$2	Units produced	0	0	1	0	1E+30
\$I\$2	Units produced	0	0	1	0	1E+30
\$J\$2	Units produced	0	0	1	0	1E+30
\$K\$2	Units produced	0	0	1	0	1E+30
\$L\$2	Units produced	0	0	1	0	1E+30
\$M\$2	Units produced	0	0	1	0	1E+30
\$N\$2	Units produced	0	0	1	0	1E+30
\$O\$2	Units produced	549	0	1	1E+30	1
\$P\$2	Units produced	11	0	1	1E+30	1
\$Q\$2	Units produced	457	0	1	1E+30	1
\$R\$2	Units produced	8	0	1	1E+30	1
\$S\$2	Units produced	111	0	1	1E+30	1
\$T\$2	Units produced	0	0	1	0	1E+30
\$U\$2	Units produced	0	0	0	0	1E+30

Variable Cells: Active plants run at full capacity with zero reduced cost, meaning the current production mix is optimal. Plants with an allowable decrease of 0 are the most sensitive any reduction in their capacity immediately changes the solution.

Constraints						
Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
\$V\$11	Could be per 100 units???	915	0	1070	1E+30	155
\$V\$12	Could be per 100 units???	138	1	138	752	138
\$V\$13	Could be per 100 units???	880	0	1013	1E+30	133
\$V\$14	Could be per 100 units???	554	1	554	336	554
\$V\$15	Could be per 100 units???	385	1	385	471	385
\$V\$16	Could be per 100 units???	49	1	49	471	49
\$V\$17	Could be per 100 units???	0	0	265	1E+30	265
\$V\$18	Could be per 100 units???	0	0	14	1E+30	14
\$V\$19	Could be per 100 units???	0	0	11	1E+30	11
\$V\$20	Could be per 100 units???	0	0	118	1E+30	118
\$V\$21	Could be per 100 units???	0	0	332	1E+30	332
\$V\$22	Could be per 100 units???	0	0	209	1E+30	209
\$V\$23	Could be per 100 units???	0	0	490	1E+30	490
\$V\$24	Could be per 100 units???	549	1	549	346	549
\$V\$25	Could be per 100 units???	11	1	11	79	11
\$V\$26	Could be per 100 units???	457	1	457	1734	457
\$V\$27	Could be per 100 units???	8	1	8	75	8
\$V\$28	Could be per 100 units???	111	1	111	89	111
\$V\$29	Could be per 100 units???	0	0	7	1E+30	7
\$V\$30	Total Capacities Could be per 100 units?	4057	0	5791	1E+30	1734
\$V\$32	Could be per 100 units???	915	1	915	155	915
\$V\$33	Could be per 100 units???	138	0	890	1E+30	752
\$V\$34	Could be per 100 units???	880	1	880	133	880
\$V\$35	Could be per 100 units???	554	0	890	1E+30	336
\$V\$36	Could be per 100 units???	434	0	905	1E+30	471
\$V\$37	Could be per 100 units???	549	0	895	1E+30	346
\$V\$38	Could be per 100 units???	11	0	90	1E+30	79
\$V\$39	Could be per 100 units???	8	0	83	1E+30	75
\$V\$40	Could be per 100 units???	111	0	200	1E+30	89
\$V\$41	Could be per 100 units???	0	0	0	0	1E+30

Constraints: Shadow prices of 1 on binding constraints confirm that every unit of added plant or port capacity translates directly into more output. PORT04 and PORT02 are the only ports running at 100% utilization and are the highest-volume bottlenecks in the network. Any demand spike past their thresholds forces a complete re-optimization of the entire network.

5 Major Findings

1

Port-Constrained, Not Plant-Constrained

PORT02 & PORT04 at 100% block 1,446 units/day. Plant investment alone won't help.

2

Cheapest Plant (PLANT07) Produces Zero

\$0.47/unit cheaper than every active plant idle because PORT02 is full.

3

Production Cost Dominates at 64%

\$296K of \$466K is production. Cost savings come from plant swaps, not freight deals.

4

PORT04 Gives 2.8× More Throughput

PORT04 expansion unlocks 1,063 units/day. PORT02 unlocks 383. Prioritize PORT04.

5

Zero Buffer at the Two Critical Ports

PORT02 (+155 units) and PORT04 (+133 units) any surge forces full re-optimization.

Course Of Action:

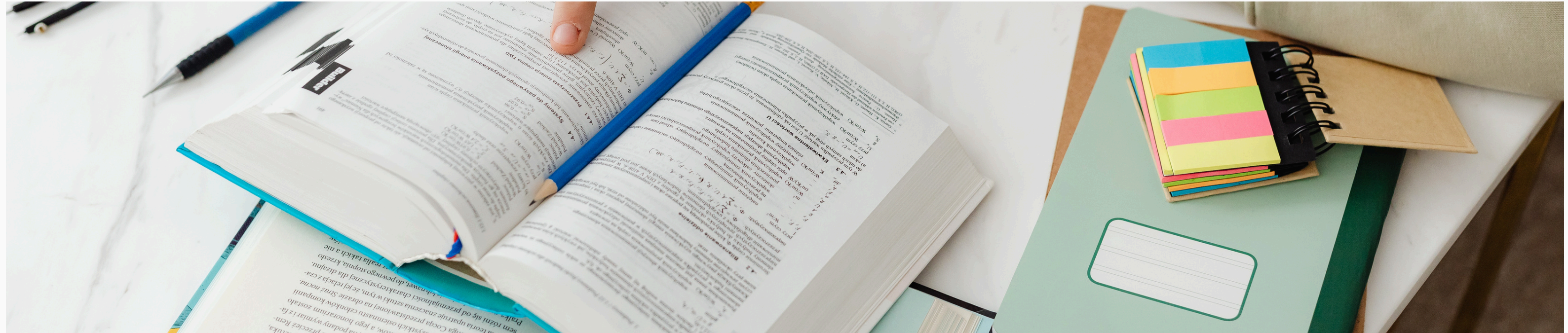
- Expand PORT04 first (+1,063 units/day), then PORT02 to unlock PLANT07.
- Fix the ports, then phase out PLANT15, 16 & 18.

LESSONS LEARNED:

Technical

- LP turned a vague cost problem into a clear mathematical model.
- Production cost dominates, so plant choice matters more than routes.
- PORT04's constraint drives much of the network behavior.
- Sensitivity report show where extra capacity is most valuable.





LESSONS LEARNED:

Process & Team

- Iterative modeling > trying to build a perfect model first.
- Data cleaning and consistent units were critical.
- Clear role-splitting made the work faster and cleaner.
- Check-ins caught constraint and cost mistakes early.

CONCLUSION

CONCLUSIONS

Implications & Future Work

- Optimal plan cuts daily cost while staying within capacities.
- High-capacity, low-cost plants carry most of the load.
- PORT04 is a key bottleneck and a good target for expansion.
- The model is a reusable baseline for future planning problems.

DS806.01

Thank You

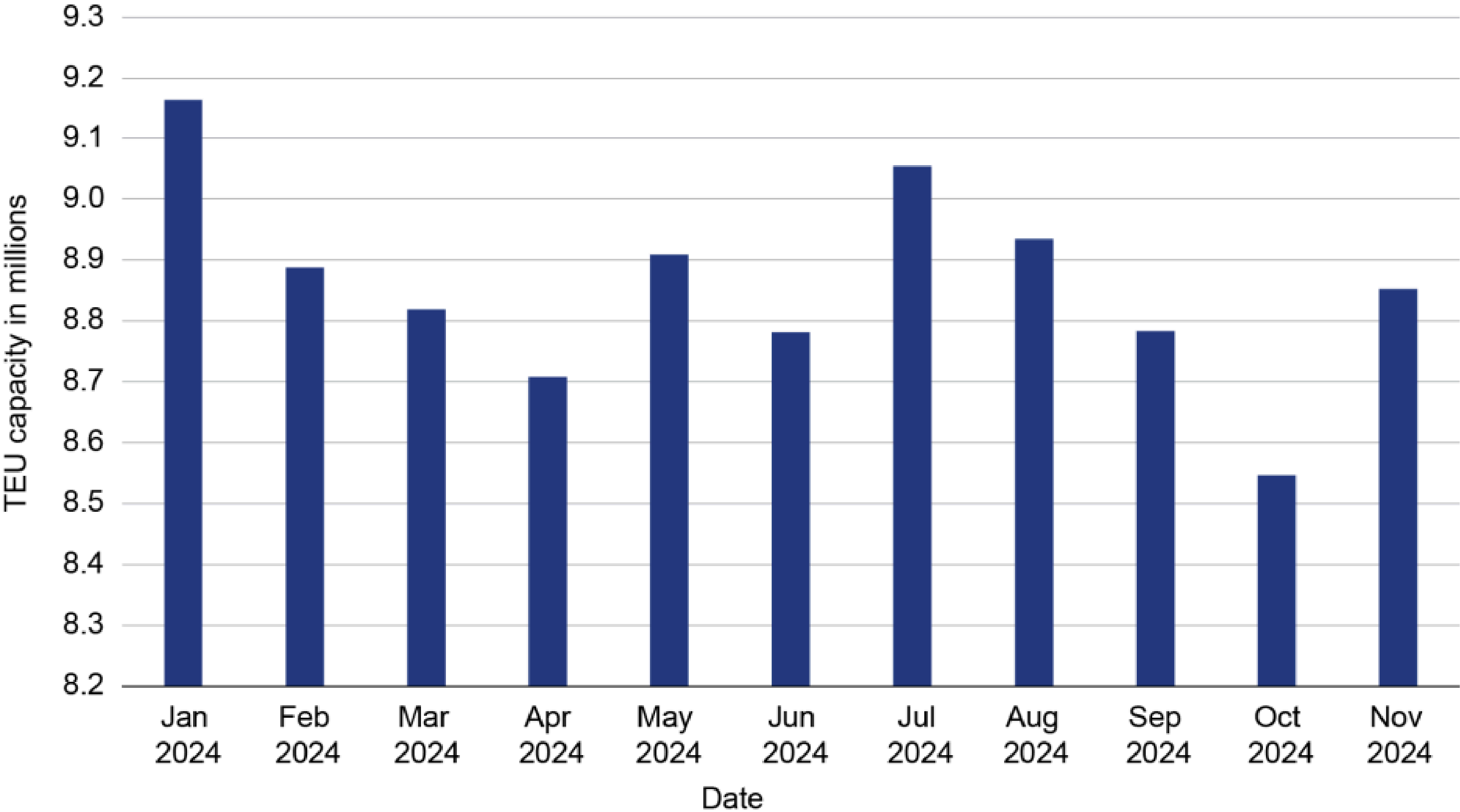
Q U E S T I O N S . . . ?

Dataset:

Supply Chain Data, by Laurin Brechter

Link: <https://www.kaggle.com/datasets/laurinbrechter/supply-chain-data/data>

Figure 22. Total Monthly Containership Capacity Serving U.S. Ports, January 2024–November 2024



Source: U.S. Department of Transportation, Maritime Administration, Office of Policy & Plans analysis of data from U.S. Customs & Border Protection, Vessel Entrance and Clearance System, and Lloyd's Register of Ships (S&P Global), available at [Latest Supply Chain Indicators \(bts.gov\)](https://bts.gov/Latest-Supply-Chain-Indicators) as of December 2024.